

Fundamental theorem for f.s. modules over PIDs

Recall the proof for f.s. abelian group, (Existence part):

(1) G : f.s. abelian group

$(\Leftarrow)$ $\exists$ surjective homomorphism

$$\mathbb{Z}^m \longrightarrow G$$

(2) Use factoring triangle

$$\Rightarrow \mathbb{Z}^m / H \xrightarrow{\cong} G$$

$$H \leq \mathbb{Z}^m$$

$$(3) \quad \mathbb{Z}^n \xrightarrow{\sim} H \quad \exists n \leq m$$

(4) Identify G with

$$\mathbb{Z}^m / \text{im } A$$

$$A \in M_{m \times n}(\mathbb{Z}).$$

$$(5) \quad C \in GL_n(\mathbb{Z}), \quad B \in GL_m(\mathbb{Z})$$

$$\mathbb{Z}^m / \text{im } A \cong \mathbb{Z}^m / \text{im } (BAC)$$

(6) (Smith normal form)

We can find C & B s.t.

$$BAC = \begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ & & & d_r & & \\ & & & & & 0 \end{bmatrix}$$

$$\begin{bmatrix} & & d_r & & \\ & & & \ddots & \\ & 0 & & & 0 \end{bmatrix}$$

$$d_1 | d_2 | \dots | d_r \dots$$

$$\textcircled{7} \quad \text{im}(BAC) \cong d_1 \mathbb{Z} \times d_2 \mathbb{Z} \times \dots \times d_r \mathbb{Z} \times \underbrace{\{0\} \times \dots \times \{0\}}_{m-r}$$

$$\mathbb{Z}^m \cong \underbrace{\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}}_{r \text{ copies}} \times \underbrace{\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}}_{m-r \text{ copies}}$$

↓

$$\mathbb{Z}^m / \text{im}(BAC) \cong \mathbb{Z} / d_1 \mathbb{Z} \times \dots \times \mathbb{Z} / d_r \mathbb{Z} \times \mathbb{Z}^{m-r}$$

Proposition (1)

$N: R\text{-module}$

$$\left\{ \begin{array}{c} R\text{-linear maps} \\ R^m \longrightarrow N \end{array} \right\}$$

$$(f: R^m \rightarrow N)$$

$\downarrow$

$$(f(e_1), \dots, f(e_m))$$

$$(f_{\underline{n}}: (r_1, \dots, r_m) \mapsto r_1 n_1 + \dots + r_m n_m)$$

$\downarrow$

$$(n_1, \dots, n_m) = \underline{n}$$

$$\left\{ (n_1, \dots, n_m) \in \underbrace{N \times \dots \times N}_{m\text{-times}} \right\}$$

In this case,

$$\text{im } f_{\underline{n}} = \left\{ \sum_{i=1}^m r_i n_i : r_i \in R \right\}$$

R -linear combos
of $n_1, \dots, n_m$

$$\begin{array}{c} i\text{-th} \\ \downarrow \\ e_i = (0, \dots, 1, 0, \dots) \\ \in R^m \end{array}$$

Corollary:

If $X = \{n_1, \dots, n_m\} \subseteq N$
is a finite subset that
generates N as an R -module

i.e. $N = \left\{ \sum_{i=1}^m r_i n_i : r_i \in R \right\}$

then $f_n : R^m \longrightarrow N$
 $(r_1, \dots, r_m) \longmapsto \sum_{i=1}^m r_i n_i$

is surjective.

$$\Rightarrow R^m / \ker f_n \xrightarrow{\cong} N$$

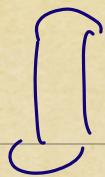
So every f.s. R -module is
a quotient R -module of R^m

for some $m \geq 1$.

Next step: Analyze R -submodules
of R^m .

Observation:

$N \leq R$: R -submodule



$N \leq R$: ideal



$N = (r)$, $r \in R$

(a) $r = 0 \Leftrightarrow N = \{0\}$

Two possibilities:

(b) $r \neq 0$

In case (1)

$\varphi: R \xrightarrow{\cong} N$: isomorphism
of R -modules

$$1 \mapsto r$$

$$s \mapsto sr = rs$$

$$\begin{aligned} s_1 + s_2 &\mapsto r \cdot (s_1 + s_2) = \varphi(s_1 + s_2) \\ &\quad \parallel \quad \parallel \checkmark \\ r \cdot s_1 + r \cdot s_2 &= \varphi(s_1) + \varphi(s_2) \end{aligned}$$

$$\begin{aligned} s_1 s_2 &\mapsto r \cdot (s_1 s_2) = \varphi(s_1 s_2) \\ &\quad \parallel \quad \parallel \checkmark \\ (r s_1) s_2 &\quad \parallel \\ &\quad \parallel \\ s_1 (r s_2) &= s_1 \varphi(s_2) \end{aligned}$$

• $\text{im } \varphi = N$, because $N = \{rs : s \in R\}$

• $\ker \varphi = \{s \in R : rs = 0\}$

$= (0)$ because R is an

integral domain.

Fact: R : Commutative ring

R is a PID

$(\Rightarrow)$ Every non-zero ideal of R
is isomorphic to R as
an R -module.

Proposition (2) $m \geq 1$

$N \leq R^m$: R -submodule.

Then $\exists n \leq m$ & an isomorphism
of R -modules

$$R^n \xrightarrow{\cong} N$$

Pf: By induction on m .

Base case $m=1$: Taken care of
by the above discussion.

Inductive hypothesis:

$$N' \leq R^{m-1} \Rightarrow \exists R^{n'} \xrightarrow{m} N' \\ n' \leq m-1$$

Inductive step:

$$N \leq R^m$$

$R^{n'}$

$n' \leq m-1$ (Inductive hypothesis)

$$N' = N \cap R^{m-1} \leq R^{m-1} = \{(r_1, \dots, r_{m-1}, 0)\}$$

$$N \leq R^m$$

$$N \leq R^m$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ N/N' & \leq & R^m / R^{m-1} \\ & \cong & R \end{array}$$

$$\uparrow \cong$$

$$R^k \quad (k=0,1)$$

$$R^{n'} \cong N' \leq N$$

$$\downarrow$$

$$N/N' \cong R^k$$

Lemma: In this situation,

$$N \cong \mathbb{R}^{n'+k}$$